

Department of Computer Science and Engineering

Academic Year 2021– 2022(Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE-‘B’ Section

Course Code & Title: EC8393 Fundamentals of Data Structures in C

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/CSE

Innovative Practice Description

- **Unit / Topic: Unit 4 / Graph and its Representation, Graph Traversal**
- **Course Outcome:** The student will be able to Identify and implement appropriate non-linear data structure operations for any application using C
- **Topic Learning Outcome:** TLO12.
- **Activity Chosen: Jigsaw**
- **Justification:**
 - The concept of graph and its representation and graph traversal is an important topic, it can be used to solve real world problem. The activity ‘JIGSAW’ is used to recall the concepts learnt and enhance the learning through this collaboration.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**

The topic graph and its representation and graph traversal taught in the class.

 - The class divided into 7 teams and each team with 7 members. The Group named as Unknown, Quadrapules, Hifi, Empassy, Rocks, Ak, Jamwal.
 - Each student in a group is assigned with any one topic. The course instructor delivered the instructions how to perform this activity.
 - For collaborative teamwork, the student who are assigned with the same topic(Graph Traversal) in each group are picked out and framed as a team – “**Expert Group**”. Each Expert team is allotted with one topic.
 - After they have solve the problem, the expert group is collapsed and all the students are re-grouped into their original group (initial team). Now each member in the team will be an expert in any one graph traversal. They have to share their ideas to all the other members in the group thus enabling each student to learn all the traversal. This will enable the students to get insight knowledge in this topic.
 - After that, the students were asked to discuss the answer for the questions as shown in the figure 2,3.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PSO1
CO2	3	3	2	2	2

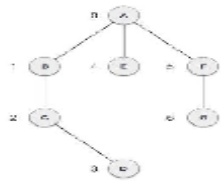
(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

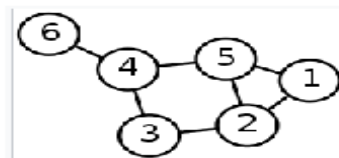
Innovative practice	PO1	PO2	PO3	PO4	PSO1
	3	3	2	2	2
Justification for correlation	The basic knowledge of engineering fundamentals is necessary to understand the concept of expression evaluation	The student should identify analyze problem to evaluate expression	The student will be able to design and find the solution for the problem		The student will be able to apply stack application in the field of information and Communication Technology

• Images / Screenshot of the practice:

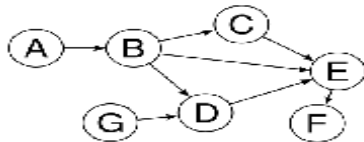
Graph Example 1



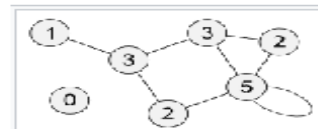
Example 2



Example 3



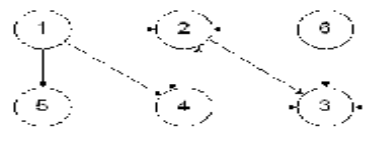
Example 4



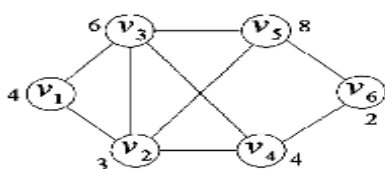
Example 5



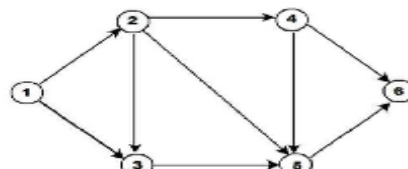
Example 6



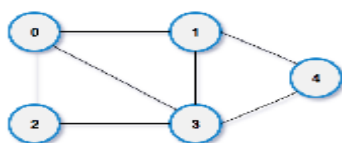
Example 7



Example 8



Example 9



Example 10

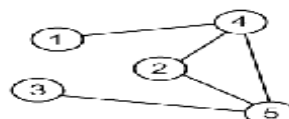


Figure 1 Questions given to students

Screen shots



Figure 2. Snapshot



Figure 3. Snapshot

- **Reflective Critique:**

- **Feedback of practice from students and other stakeholders:**

- Most of the students enjoyed the activity because this is the first time experience.
 - They felt that the self-learning skills improved due to this type of activity.
 - These type of activities are interesting which involve us to learn and teach the concepts to our group members.
 - It helps them to improve the communication skills i.e interacting effectively with the group and summarizing the concept.

- **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Through this activity the students are **made to work in a team and share their ideas** to others and also **improve their communication skills**. Thus the students were made to understand that the team work will give a better solution

❖ ***Challenges faced in implementation:***

- **Time management:** while discussing the concepts and teaching to the other group members the time allotted was not enough.

References:

- <https://www.ritrjpm.ac.in/images/computer-science/>

Signature of Faculty Member

HOD